GENAI BASICS • LESSON 04 • MODULE 2: MULTIMODALITY

Image Generation

Services, Prompt Techniques, and Control Patterns

ChatGPT, Nano Banana, Leonardo.ai, Midjourney, Kling, and FLUX. From an interactive quiz and the market map to e-commerce packshots, edits by instruction, and a character who stays recognizable in every scene.



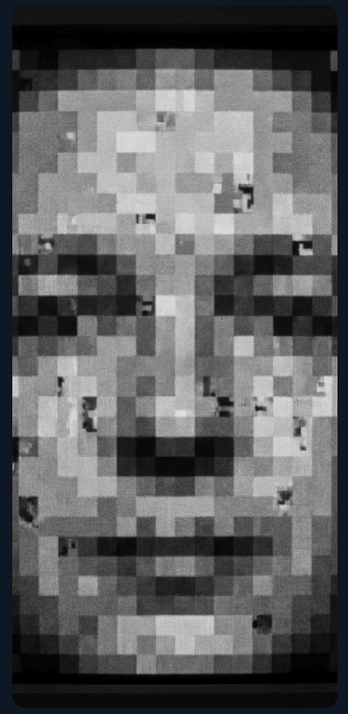


INTERACTIVE HOOK • WARM-UP QUIZ

Guess: Which is a real human, and which is AI?

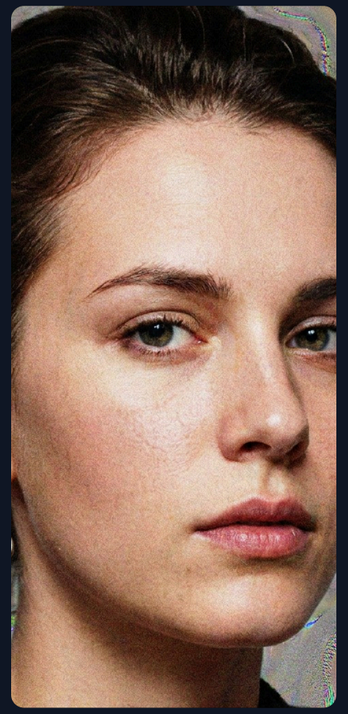
Six portrait images. Post the letters of those you believe are genuine photographs vs. AI generated in chat.

[A]



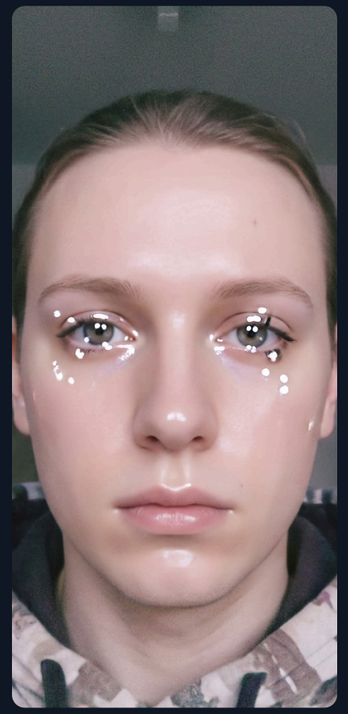
Portrait A

[B]



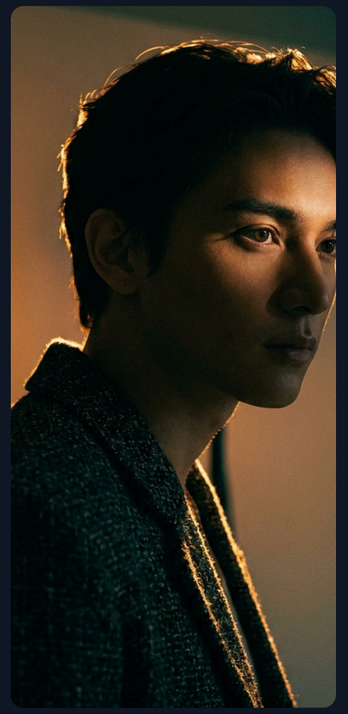
Portrait B

[C]



Portrait C

[D]



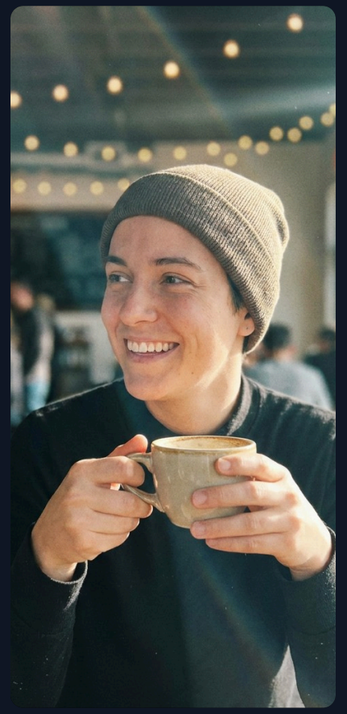
Portrait D

[E]



Portrait E

[F]



Portrait F

Chat Poll: Enter your guesses. For example: "A and B are AI, E and F are real" or "Only F is real". What visual clues are you looking at?

THE REVEAL • 2014-2025

All 6 are AI generated. A decade of evolution

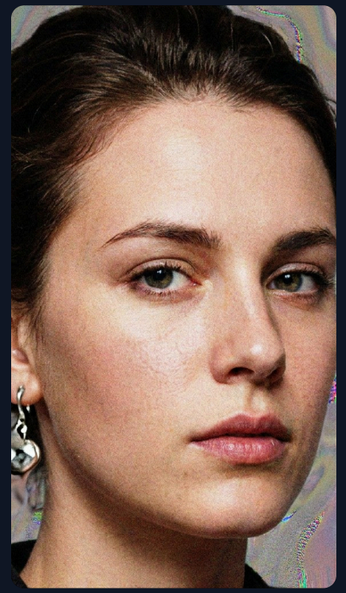
Not a single real photograph. A reconstruction of each era's style (all frames generated in 2026): how generative models crossed the uncanny valley.

2014 GAN



Goodfellow GAN
64×64 px grayscale noise blobs. The birth of deep generative models.

2018 StyleGAN



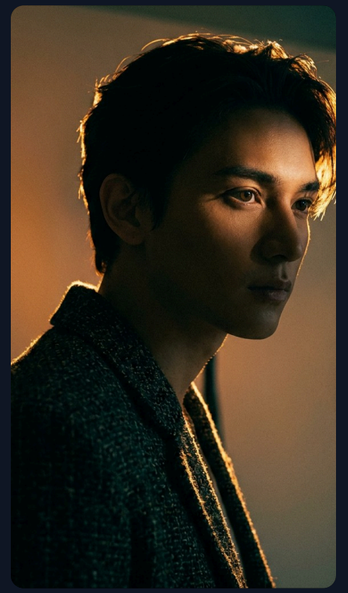
NVIDIA StyleGAN
Early photorealism, but melting earrings, ears, and warped backgrounds.

2022 SD 1.5



Stable Diffusion
Open-source diffusion explosion. Glossy, plastic skin and vacant gaze.

2023 MJ v5



Midjourney v5
Cinematic aesthetic victory. Studio lighting, dramatic chiaroscuro.

2024 FLUX.1



FLUX.1 Dev/Pro
Authentic skin pores, micro-imperfections, natural teeth and anatomy.

2025 NB Pro



Nano Banana Pro
Smartphone optics, bokeh, sensor grain. Indistinguishable by eye.

> **Engineering Takeaway:** Modern models generate beauty effortlessly. The central problem of 2026 is **controllability**: how to steer models to generate exact specifications and maintain subject identity across 10 scenes.

COURSE ROADMAP

Module 1 Completed. Entering Pixels

01 **Foundations & API**
L01–L03

02 **Multimodality**
L04 Images
L05 Video • L06 Audio

03 **AI assistants at work**
L07–L09

04 **Open models**
L10

05 **Agents & MCP**
L11–L13

06 **Idea to product**
L14–L16

> **Carried over from Module 1:** Prompt engineering principles from L02 apply directly to vision models. Specificity beats buzzwords, slot separation prevents collision. But instead of text few-shots, we now supply **visual reference images**.

ARCHITECTURE • INDUSTRY OVERVIEW

Three Layers of the Image Generation Ecosystem

Model ≠ service. The same model can live in an API, in a creative studio, and in a chat.

LAYER 3

Multimodal Chat

The model reasons in text, draws, and remembers the dialogue. Best place for edits and in-image text.

- **ChatGPT** — GPT Image 2.5
- **Gemini** — Nano Banana 2 / Pro
- **Grok** — Grok Imagine 2.0 (Aurora model)

▲ build on models and APIs from the layer below ▲

LAYER 2

Creative Studios

Canvas, masks, layers, upscaling, presets. Own models and/or third-party models via API.

- **Leonardo.ai** — own Phoenix / Lucid + third-party
- **Kling** — own IMAGE 3.0 / O1 → video
- **Midjourney** — own V8.2, Web / Discord only
- **Krea, Sintx** — aggregate third-party models

▲ build on models and APIs from the layer below ▲

LAYER 1

Foundation Models & APIs

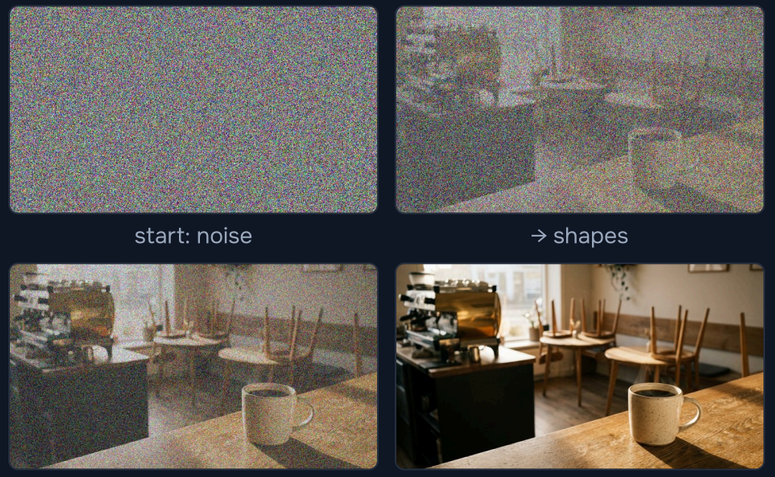
Train the weights, run the GPUs, and expose REST / Python APIs for automation.

- **OpenAI** — GPT Image 2 / 2.5
- **Google** — Nano Banana: Gemini API, Vertex
- **Black Forest Labs** — FLUX.2 / FLUX 3
- **Stability AI** — SD 3.5 / SDXL
- **ByteDance** — Seedream 5.0

service on its own models aggregator of third-party models One vendor can sit on several layers: Google is both an API (Layer 1) and the Gemini chat (Layer 3).

How Models Draw: Diffusion vs Autoregression

Diffusion: from noise, everywhere at once

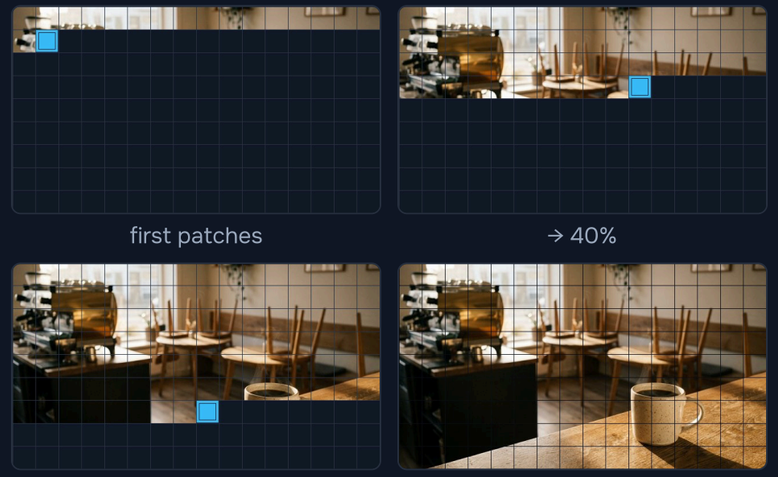


start: noise → shapes → details → final

- Starts from pure noise and removes it over 20–50 steps **across the whole image at once**; the prompt steers every step.
- Different **seed** = different noise = different image. Hence the studio knobs: **seed, guidance, steps, negative prompt**.

Who: Stable Diffusion / SDXL, FLUX, Midjourney, most studio models.

Autoregression: patch by patch, like text



first patches → 40% → 70% → done

- An image is a sequence of **patch tokens**; the model predicts the next one, like the next word in a sentence.
- The same multimodal LLM that reads the prompt does the drawing → strong at **text, instructions and edits**. Often a hybrid.

Who: GPT Image, Grok Aurora; Nano Banana — architecture not disclosed.

In practice: diffusion — sliders and seed; chat models — prompt as a spec and edits in words. The techniques in this lesson work for both.

LAYER 1 • FOUNDATION PROVIDERS

Who Trains Foundation Models & Exposes APIs

The same providers you already called in L03 via the SDK also expose image generation endpoints.

GOOGLE

Nano Banana 2 / Pro / 2 Lite

`gemini-3.1-flash-image`, `gemini-3-pro-image`, `gemini-3.1-flash-lite-image`.
Strengths: speed, world knowledge and Search grounding, multi-reference input, same SDK as L03.

OPENAI

GPT Image 2 / 2.5

`gpt-image-2`, `gpt-image-2.5-flare` (fast), `gpt-image-2.5-sunburst` (precise edits).
Strengths: in-image text, strict instruction following, conversational edits.

BLACK FOREST LABS

FLUX.2 (Pro / Flex / Dev) → FLUX 3

API: BFL, Replicate, fal.ai. FLUX.2 Dev / Klein are open weights, up to 10 references. FLUX 3 (07.2026) is multimodal.
Strengths: photorealism, hex color control, local hosting.

STABILITY AI

Stable Diffusion 3.5 / SDXL

Open-weights ecosystem staple. API access and local hosting (L10).
Strengths: thousands of community fine-tunes and style LoRAs.

MIDJOURNEY

Midjourney V8.2 + Edit Model

Web and Discord only, no public API. Edit Model (08.2026): edits, inpaint / outpaint, up to 4 references.
Strengths: aesthetics and cinematic style.

PROGRAMMATIC ACCESS

Direct API Calls

`POST /v1/images/generations` or `client.models.generate_content`.
Cost: ≈ $0.02–0.06 per 1K image on fast models; Pro and 4K cost more. Latency: 3–15 s.

MARKET • OCTOBER 2026

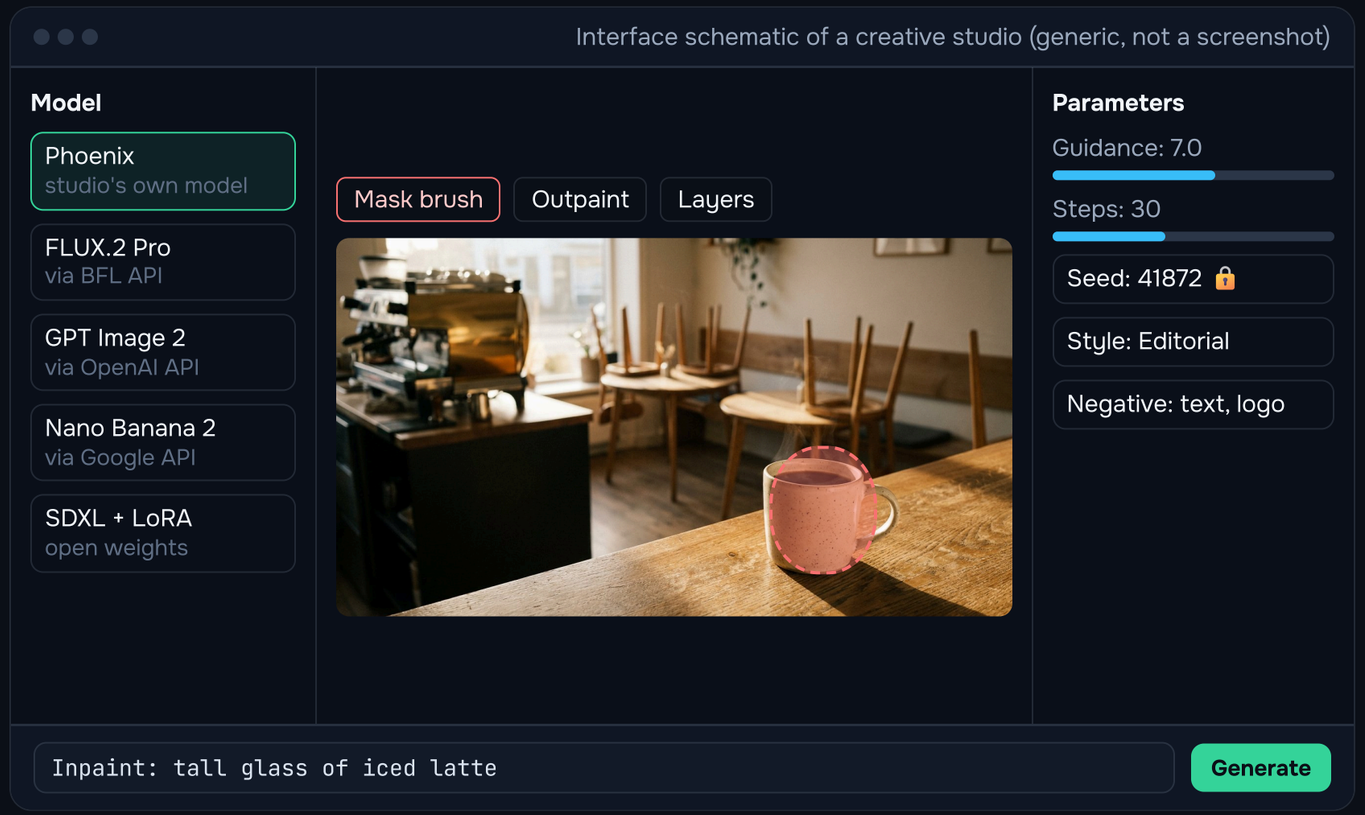
The Image Generation Market: October 2026

MODEL · VENDOR	STRENGTH	REFERENCES	ACCESS & PRICE
GPT Image 2 / 2.5 OpenAI	Text and infographics, instruction following, conversational edits; #1 Arena	up to 16 (2.5; sources disagree)	ChatGPT (incl. Free); API ≈ $30 / 1M image output tokens
Nano Banana 2 / Pro / 2 Lite Google	Speed, world knowledge and Search, 4K, in-image text translation	NB 2: up to 14 objects + 5 characters	Gemini app; API: 2 Lite ≈ $0.034 / 1K image
Grok Imagine Image 2.0 xAI / SpaceXAI	Magic Wand selection, templates (product, headshot); #2 Arena	up to 5	Grok app; API ≈ $0.02–0.06
Midjourney V8.2 Midjourney	Aesthetics and style: `--sref`, moodboards, personalization	Edit Model: up to 4	$10–120 / month; no API
FLUX.2 / FLUX 3 Black Forest Labs	Open weights (Dev, Klein), 4 MP, exact hex colors	up to 10	API: BFL / fal / Replicate; local (L10)
Kling IMAGE 3.0 / O1 Kuaishou	Cinematic framing, image series, direct bridge to Kling video	up to 10	kling.ai (credits), API
Seedream 5.0 ByteDance	4K, Search in Lite; Pro splits a render into PNG layers	up to 10 (Flash)	API, partner platforms
Ideogram 4.0 Ideogram	Typography; official **JSON prompts**; open weights 9.3B	—	ideogram.ai; Apache 2.0 weights
Recraft V4 Recraft	Native editable **SVG** output, brand styles	style sets	recraft.ai, API

> **Arena, 08.2026:** GPT-Image-2 is #1, Grok Imagine 2.0 is #2 (Text-to-Image and Image Edit). Limits and prices change monthly — re-check before class.

Why Use a Creative Studio Instead of a Raw API?

Leonardo.ai, Kling, Krea, Sintx: canvas, masks, and presets on top of their own and third-party models.



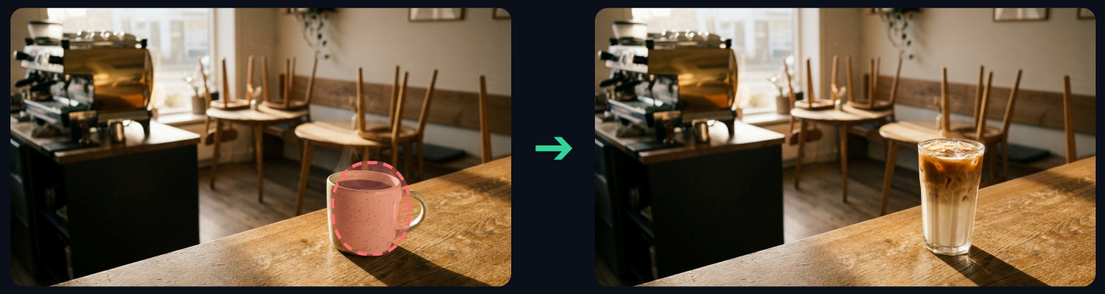
- **Own + third-party models:** Leonardo (Phoenix, Lucid) and Kling (IMAGE 3.0) run their own; FLUX, GPT Image, Nano Banana, SDXL sit alongside — one subscription instead of five API keys.
- **Canvas:** layers, brush masks (inpainting), extending the frame (outpainting).
- **Presets & LoRAs:** brand style or "product shoot" in one click.
- **Control knobs:** guidance, seed, ControlNet by pose or edges.

> **Definition:** a creative studio is a visual shell over models (its own and third-party) that turns generation into a workflow.

Canvas & Sliders vs Editing in Words

One task — swap the mug for an iced latte — in two paradigms.

Schematic: creative studio (e.g. Leonardo)



brush mask | prompt for the area | negative prompt | seed 41872 🔒

Creative studio: a graphics editor

You drive the process: paint a mask over the mug (inpainting), write a prompt for that area, lock seed and guidance. Pixel-level control at the cost of manual work.

Dialogue schematic: ChatGPT / Gemini



You

Replace **only** the mug with a tall glass of iced latte. Keep the light, counter and background.



Model

Done: mug replaced, everything else unchanged.

Chat model: talking to a director

No mask needed: the model finds the object from the description (edit by instruction). Fast, but precision depends on your wording — and on checking the result.

ENGINEERING EVALUATION • SELECTION MATRIX

Selection Matrix: Studio vs Chat vs API

CRITERION	CREATIVE STUDIO (LEONARDO, KREA)	MULTIMODAL CHAT (CHATGPT, GEMINI)	API (PYTHON SDK / VERTEX / FAL)
Entry barrier	Moderate (interface, sliders, layers)	Minimal (plain language)	High (code, keys, pipelines)
Targeted edits	Brush mask / canvas (inpainting)	Edit by instruction (in words)	Mask in the request or edit by instruction
In-image text	Depends on the selected model	Excellent (GPT Image 2.x, Nano Banana)	Excellent (GPT Image 2.x, Nano Banana, FLUX.2)
Character consistency	Character reference, seed lock	Multi-reference (Nano Banana 2: up to 14 objects + 5 characters)	References in request `contents`
Batch (500+ items)	Impractical (clicks, credit limits)	Impossible (manual chat)	Optimal: script + DB. Per image: Nano Banana 2 Lite ≈ $0.034 (1K), legacy Nano Banana ≈ $0.039, Grok ≈ $0.02–0.06

> **Engineering rule:** concepts and quick text ➔ **chat**. Commercial art with a hand mask ➔ **studio**. Mass banner generation from a catalog ➔ **API script**.

Six Words — Model Makes Every Creative Decision



COMPLETE VAGUE PROMPT

```
a cozy coffee shop in the morning
```

Morning? Outside it's raining. Strangers, a cat, and a fireplace were never requested — the model filled empty latent space with training data averages.

Six Slots — What Controls What

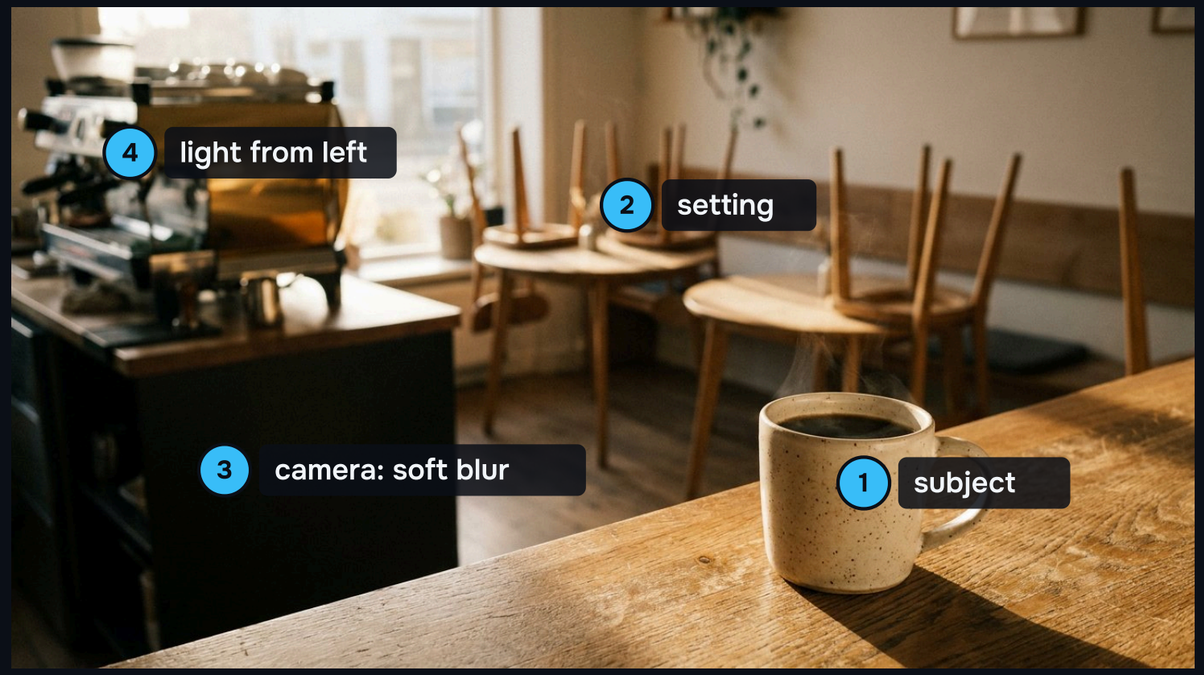
RUNNABLE SLOT PROMPT (COPY & RUN) — high scene control

```
Subject: a speckled cream ceramic mug of black coffee with steam,
on worn oak counter in right third.
Setting: empty Scandinavian coffee shop at 7 a.m., chairs up on
tables, brass espresso machine on left.
Camera: eye-level, 35mm lens, f/2.0, shallow depth of field.
Lighting: low warm morning sunlight from left window, long soft
shadows.
Style: realistic editorial photography.
Constraints: no text, no logos.
```

NEGATIVES DEPEND ON THE MODEL

Gemini and GPT follow positive phrasing better: *"empty coffee shop"* instead of *"no people"*. Negations via `--no` / negative prompt are for **Midjourney and SD / FLUX** in studios.

> Instead of "cozy, 4k, masterpiece" — physical details: time, optics, light.



1 Subject — main object, texture, placement in frame (right third).

2 Setting — time and context: "7 a.m., chairs up" removes random people.

3 Camera — 35mm (natural, moderately wide angle), f/2.0 — soft bokeh.

4 Lighting — low light from the left adds volume and long shadows.

5 Style — editorial photo instead of a glossy 3D render.

6 Constraints — what to exclude: text, logos.

Critical Details & Text as an Explicit Block




```
Must-have:
• chalkboard sign with text exactly "FLAT WHITE 3.50" and "OPEN 7:00"
• one fresh croissant behind the mug
• small glass vase with lavender to the left of the mug
• no other text
```

ACCEPTANCE CHECKLIST FOR FRAME A3

- ✅ **Text** "FLAT WHITE 3.50" and "OPEN 7:00" — no typos
- ✅ **One croissant** behind the mug
- ⚠️ **Lavender** is there, but on the right, not the left
- ❌ **Extra text:** menu board at the top

> **Rules:** text in quotes with the word `exactly`; quantities as numbers; positions stated explicitly. Failed → **targeted edit**, not a reroll.

Frame A3 — the result of the prompt on the right. Check it against a checklist, not "by eye".

One Subject — Six Lens and Light Coordinates



`low angle` heroic perspective



`top-down` flat lay arrangement



`24mm wide` subject + environment



`85mm f/1.8` portrait optical bokeh



`golden hour` warm backlight



`softbox studio` balanced commercial light

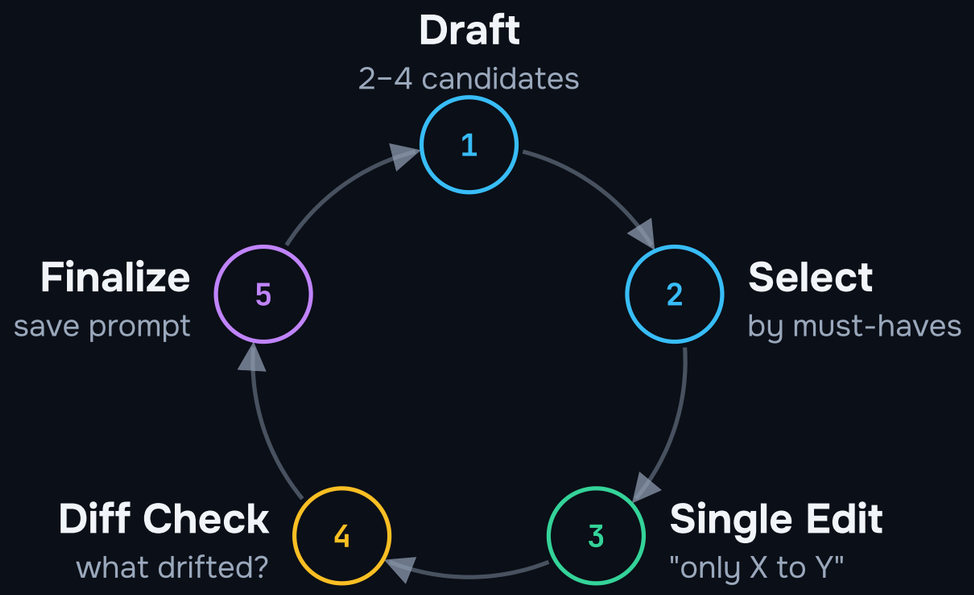
LECTURE EQUATOR

5-Minute Break

05:00

Coming next: targeted edits by instruction, a consistent courier character, and a commercial product photoshoot.

Iterate in Loops — Never Reroll the Whole Dice



ENGINEERING REFINEMENT TEMPLATE

```
Edit this image. Modify only object X to Y.
Preserve: angle, lighting, background, and shadows.
Do not change: surrounding room geometry.
```

- **One edit per step** (equivalent to an atomic git commit).
- **Explicitly declare protected regions** to preserve context.
- **After ~5–8 edits** drift becomes visible — start a fresh chat from the best frame.

One Master Frame — Three Targeted Edits



Original A2: morning coffee shop, mug



Object (B1): mug → iced latte



Lighting (B2): rainy evening outside



Style (E2): flat vector illustration

FORMULA **Frame A2 + one edit per step**

"Replace only X with Y. Keep: camera angle, lighting, background, geometry."

PROMPT B1: OBJECT SWAP

```
Edit this photo. Replace only the ceramic mug with a tall glass of
iced latte with visible coffee and milk layers and ice cubes. Keep
everything else exactly the same: camera angle, counter, espresso
machine, background, light direction and shadows.
```

PROMPT B2: LIGHTING AND WEATHER

```
Edit this photo. Change only the time of day to a rainy evening: dark
dusk outside window, raindrops on glass, warm interior pendant lamps
switched on. Keep exact same composition, objects, mug and camera
angle.
```

> **ChatGPT / Gemini / Kling:** edit by instruction — no mask needed, the model finds the mug itself.
> **Leonardo:** inpainting — paint a mask over the mug + a prompt for that area.

Courier Master Sheet: Multi-Angle Formula

A naive prompt draws a new person in every scene. The engineering approach: first 4 angles on grey and physical anchors.



Character sheet: 4 angles, neutral pose, studio light

5 fixed anchors: short curly hair • small silver hoop earring • bright yellow jacket with grey stripe • turquoise backpack • white sneakers

NAIVE **"food courier cycling in the rain"**

Every scene gets a new face, age and jacket. No consistency.

ENGINEERING **Character sheet: 4 angles on grey**

Studio light, full body: front, 3/4, profile, back. Master reference for every scene.

MASTER SHEET PROMPT (C1)

```
Character reference sheet on a plain light grey background: the same young woman
food delivery courier shown four times in a row - front view, three-quarter view,
side profile, back view - full body, standing neutral pose.
Fixed appearance: about 28 years old, short curly dark-brown hair, light-brown
skin, small silver hoop earring in left ear, bright yellow waterproof jacket with
reflective grey stripe across chest, black cargo trousers, white sneakers, square
turquoise thermal backpack. Studio photography, clean even lighting. No brand
names, no logos, no text.
```

Where to use it: Gemini, ChatGPT, Leonardo, Kling. In Midjourney — via the **Edit Model** (up to 4 references; replaced `--cref` / `--oref` in 08.2026). Save as master `C1_sheet.png`.

Any Person as Reference: From 2–3 Photos to a Master Sheet

INPUT 1 • REFERENCES — 2–3 real face photos



Front, studio



Emotion, light



Daylight

INPUT 2 • PROMPT (GEMINI / CHATGPT / LEONARDO) — **holds the face**

```
Create a photorealistic character reference sheet based on the attached
photos of Ihar Rubanovich. Same adult man, not a generic model.
Preserve facial identity: face shape, eyes, nose, lips, jawline,
hairline, skin tone. Do not beautify, rejuvenate or stylize his face.
Four angles: front, three-quarter, side profile, back. Full body, neutral
pose.
Outfit: yellow courier jacket with grey stripe, black cargo trousers,
white sneakers, turquoise thermal backpack. Light grey studio background.
No text, no logos.
```

What to send: photos + prompt in the chat; in code — files in `contents`. Only your own photos or with the person's consent.



Result: the face is recognizable in all 4 angles. Check by eye: small features can drift.

ANTI-PATTERN

Without `do not beautify` the model smooths the skin and "improves" the face into a stock one.

BRIDGE TO PRODUCTION

Attach the sheet as a character reference to scenes and to Kling / Veo for video.

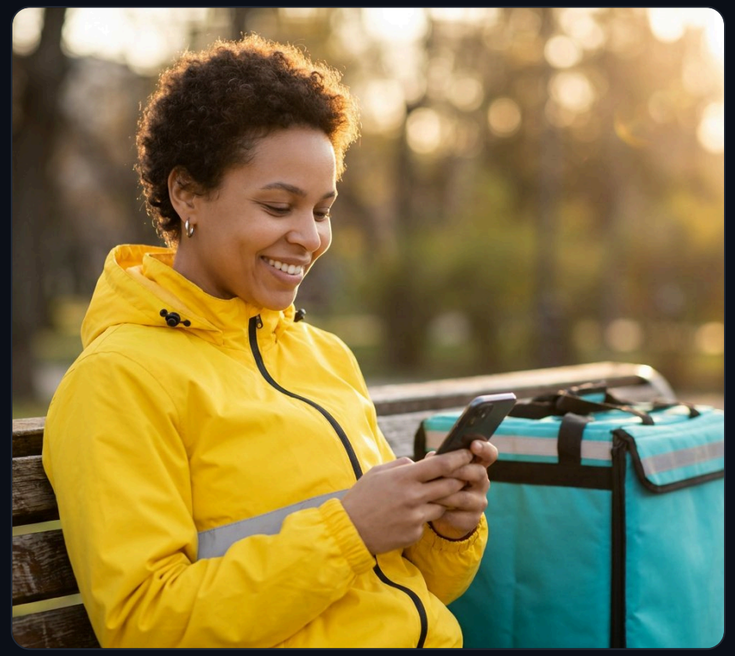
Courier in Scenes & the First Video Frame (Kling / Veo)



Scene 1 (16:9): cycling in the rain ➔ **base for video**



Scene 2: doorstep handover (35mm)



Scene 3: park bench break (85mm)

SCENE TRANSFER PROMPT

```
Use attached character sheet (C1) as identity reference:
The same courier (same face, curly hair, silver hoop earring, yellow jacket
with grey stripe, turquoise backpack) is riding a city bicycle through a rainy
evening street with wet cobblestones reflecting shop lights. Medium shot, 50mm
lens, motion blur in background. Realistic photography. No logos, no text.
```

Formats: 16:9 — YouTube and landscape video, 9:16 — Reels / Shorts (format selector).

BRIDGE TO VIDEO **Pipeline to Kling 3.0 / Veo 3.1**

1. **Export frame:** `C2_rain.png` (16:9) — the first frame (t_0).
2. **Image-to-Video:** upload into the **First Frame** slot.
3. **Motion prompt:** `"courier starts pedaling, wheels splash puddles, camera tracks along"`.
4. **Result:** face and outfit are noticeably more stable than generating from scratch; still check frames for drift.

In tools: Gemini / ChatGPT (reference in chat) • Leonardo (character reference + seed) • Midjourney (Edit Model) • Kling / Veo (First Frame).

One Packshot — Four Commercial Assets



Master packshot (D0): product on pure white



Lifestyle (D1): kitchen, morning



Flat lay (D2): top-down



In hands (D3): product scale



Banner (D4): 1:1 promo poster

1. MASTER PACKSHOT PROMPT (D0)

```
Studio e-commerce packshot of a single handmade ceramic
mug: cylindrical body, speckled cream glaze with small
brown dots, unglazed terracotta-brown band at base, round
C-shaped handle on right. Centered, 3/4 view, pure white
background, soft even light. No text, no logo.
```

2. CATALOG SCENE PROMPT (WITH D0 REFERENCE)

```
Product photo session. Place exactly this mug (same shape, speckled cream glaze, terracotta base
band, handle on right) on light travertine kitchen counter next to morning window. Soft light,
50mm f/2.8. Do not change mug shape, clay glaze, or proportions.
```

Engineering control: reference `D0` + a "do not change the shape" constraint keep the mug noticeably more stable; final acceptance is done by a human.

Explicitly Assign the Role of Each Image



Image 1 — character (courier)



Image 2 — product (mug)



Result: the courier drinks from that exact mug in a morning café

ROLE RULE **Never say "combine these" (you get mush). Assign roles:**
Image 1 = identity reference (hero), Image 2 = product reference (item).

COMPOSITION PROMPT (NANO BANANA / CHATGPT)

```
Image 1 is the courier character, Image 2 is the exact ceramic mug.
The courier from Image 1, in her yellow jacket with grey stripe, sits at a coffee shop window
counter on her break and drinks from exactly the mug from Image 2 (same speckled cream glaze and
terracotta base). Turquoise backpack on floor next to stool. Morning light, 35mm lens, realistic
photography. No logos, no text.
```

Batch Generation via Code — Same SDK as L03

```python
import csv
from google import genai
from google.genai import types
from PIL import Image

client = genai.Client()
ref = Image.open("packshot.png")

for row in csv.DictReader(open("catalog.csv")):    # ~50 products
    resp = client.models.generate_content(
        model="gemini-3.1-flash-image",
        contents=[ref, PROMPT_SPEC.format(**row)],
        config=types.GenerateContentConfig(
            response_modalities=["IMAGE"],
            image_config=types.ImageConfig(aspect_ratio="16:9")))
    for part in resp.parts:          # response = list of parts
        if part.inline_data:         # keep only the image
            part.as_image().save(f"{row['sku']}.png")
```

~50 cards in ~10 min for ~$2–3

Instead of a day of manual browser work: a product CSV → a generation loop. Roughly 10 s and $0.04–0.06 per image.

> **Don't take `parts[0]` blindly:** the response may also contain text parts. Look for the part with `inline_data`.

> **10/90 in code:** the model call is 5 lines. The rest is catalog parsing, prompt templating, retries and automated artifact checks.

Technical Limitations & Production Boundaries

- **Anatomy & small text:** fingers in crowds and long paragraphs still fail.
- **Detail drift:** after ~5–8 consecutive edits, drift in faces and logos becomes visible.
- **Copyright:** in the US a purely AI-generated image is not protected (Thaler: on 02.03.2026 the Supreme Court declined to hear the case). Disney / Universal v. Midjourney is ongoing.
- **Provenance:** SynthID — an invisible watermark in the pixels (Google; OpenAI since 05.2026) + C2PA — signed metadata that is lost on a screenshot.



GREEN • AUTONOMOUS
Moodboards, concept art, internal presentation slides, storyboards.

YELLOW • HUMAN-IN-THE-LOOP
E-commerce cards, marketing banners, YouTube thumbnails. Human sign-off required.

RED • PROHIBITED
Legal contracts, medical instructions, third-party trademarks, non-consensual likeness.

PRACTICE • HOMEWORK ASSIGNMENT #4

Product Photoshoot or Delivery Courier Series

Track A: Commercial Product Photoshoot

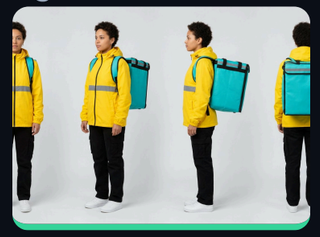
Pick a physical product on white background ➔ 4 commercial angles (lifestyle, flat lay, in hands, banner).

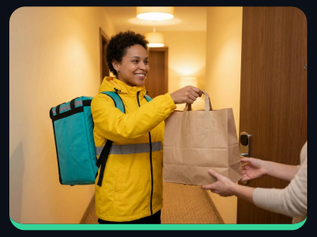





Track B: Glovo Food Delivery Courier

Character sheet ➔ 3 contextual scenes with visual anchors ➔ 1 targeted edit by instruction.






SUBMISSION REPOSITORY STRUCTURE

```
genai-homeworks/
└── L04/
    ├── images/ (01_free.png, 02_structured.png...)
    ├── prompts.md (prompts, tool, timestamp)
    └── qa.md (drift observations, what succeeded)
```

Required: Run one prompt across two tools (e.g. Leonardo vs. ChatGPT) and analyze the differences in `qa.md`. Invite collaborator: `ihar_rubanovich@epam.com`.

TAKEAWAYS • BRIDGE TO VIDEO

Four Principles of Visual AI Engineering

1. Tool to Match Task

Chat for concepts and text, studio for canvas and masks, API for batch automation.

2. Structure Over Freeform

6 explicit slots (Subject, Scene, Camera, Light, Style, Negatives) + Must-Have block.

3. Surgical Refinement Over Rerolls

Atomic "modify only X, preserve Y" commands outperform blind dice rolling.

4. Consistency via Visual Anchors

Master character sheet + 4–6 distinctive textural anchors preserve identity anywhere.



Lesson 05: Video Generation

Today's courier frames become the keyframes of tomorrow's video models (Kling, Veo, Runway). Our courier will start moving!